Algebra II with Trigonometry Chapter 11.4

Mental Math (Gemini)

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Identifying the Shift If you replace x with $x+3$ and y with $y-5$ in a conic equation, in which directions and by how many units will the graph shift?

Identifying from General Form I Look at the equation $4x^2-9y^2+16x+18y-11=0$. Based on the squared terms, does this represent a parabola, an ellipse, or a hyperbola?

Identifying from General Form II Consider the equation $x^2-6x-12y+33=0$. What type of conic section does this represent?

Identifying from General Form III What type of conic is represented by the equation $9x^2+4y^2-54x+2y+46=0$?

Center of an Ellipse What are the (x,y) coordinates for the center of the ellipse given by the equation $\frac{x^2}{4}+(y+2)^2=1$?

Center of a Hyperbola What are the (x,y) coordinates for the center of the hyperbola given by the equation $\frac{(y-1)^2}{25}-(x+3)^2=1$?

Parabola Vertex What is the vertex of the shifted parabola described by the equation $(y+1)^2=16(x-3)$?

Parabola Direction Based on its standard form, does the parabola $(x-5)^2=-8(y+2)$ open upwards, downwards, to the left, or to the right?

Ellipse Axis Length For the ellipse $\frac{(x-2)^2}{9}+\frac{(y-1)^2}{16}=1$, what is the total length of its major (longest) axis?

Reverse Engineering the Translation If a conic centered at the origin is shifted 7 units to the right, what mathematical expression will replace x in its new equation?